

Splunk Windows Service Persistence Detection Lab

Detecting Malicious Service Creation Using Windows Event ID 7045

Overview

This project documents a hands-on SOC detection lab focused on identifying malicious Windows service creation using Splunk and Windows Event Logs.

The objective of the lab was to understand how attackers abuse Windows services for:

- Persistence
- Privileged execution
- Background malware execution
- Stealthy long-term access

Using Splunk Enterprise and Windows telemetry, I investigated:

- Windows service creation
- Event ID 7045
- Service-based persistence
- Service execution behavior
- Correlation between service creation and process execution
- Detection tuning concepts

This lab introduced another major attacker persistence and execution technique commonly observed in enterprise environments.

Why Windows Services Matter

Windows services are highly valuable to attackers because they:

- Can automatically start during boot
- Often execute with SYSTEM privileges
- Run silently in the background
- Blend into legitimate operating system activity
- Persist even after reboots

Services are heavily abused in:

- Ransomware operations
- Remote execution frameworks
- Malware persistence
- Lateral movement
- Privilege escalation activity

Because legitimate administrators also create services regularly, service creation events are considered:

High-value but noisy telemetry

This makes proper investigation and correlation extremely important for SOC analysts.

Lab Environment

Virtual Machines Used

Ubuntu VM

Used as the Splunk Enterprise server.

Windows VM

Used as the endpoint telemetry source and service persistence simulation system.

Kali Linux VM

Used as part of the broader SOC lab environment.

Technologies Used

Tool	Purpose
Splunk Enterprise	SIEM and log analysis
Windows Event Logs	Security telemetry
CMD	Service creation execution
sc.exe	Windows service management utility
Windows Services	Persistence mechanism
VirtualBox / VMware	Virtualization

LAB — Malicious Service Creation

Objective

The objective of this lab was to simulate attacker persistence through Windows service creation and investigate the resulting telemetry using Splunk.

The following command was executed as Administrator:

```
sc create UpdaterService binPath= "C:\Windows\System32\notepad.exe" start= auto
```

The service was then started using:

```
sc start UpdaterService
```

What This Command Does

The command creates a Windows service named:

UpdaterService

configured to:

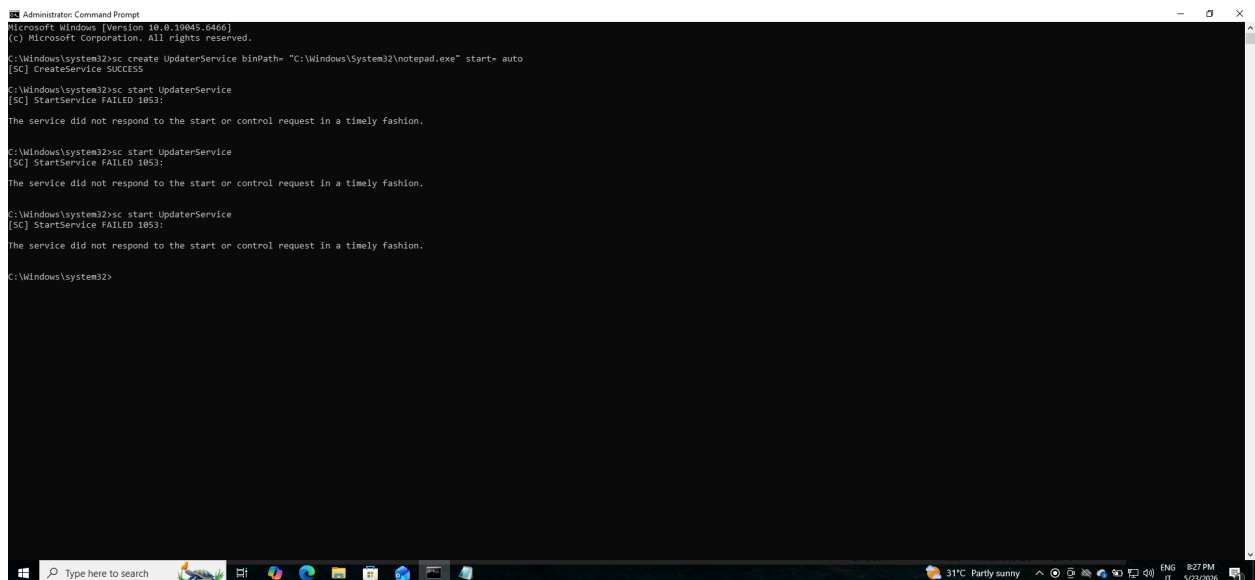
- Automatically start during boot
- Execute `notepad.exe`
- Run in the background
- Persist on the system until removed

This generated:

Windows Event ID 7045

which records service installation activity.

Screenshot 1 — Windows Service Creation Command



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.0460]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>sc create UpdaterService binPath= "C:\Windows\System32\notepad.exe" start= auto
[SC] CreateService SUCCESS

C:\Windows\system32>sc start UpdaterService
[SC] StartService FAILED 1063:

The service did not respond to the start or control request in a timely fashion.

C:\Windows\system32>sc start UpdaterService
[SC] StartService FAILED 1063:

The service did not respond to the start or control request in a timely fashion.

C:\Windows\system32>sc start UpdaterService
[SC] StartService FAILED 1063:

The service did not respond to the start or control request in a timely fashion.

C:\Windows\system32>
```

Why Attackers Abuse Services

Attackers frequently abuse Windows services because they provide:

- Long-term persistence

- Silent execution
- Elevated privileges
- Automatic execution after reboot
- Legitimate-looking system behavior

Unlike standard user processes, services operate in the background and often avoid immediate user visibility.

This makes them highly effective for:

- Ransomware execution
 - Malware persistence
 - Remote access tools (RATs)
 - Lateral movement frameworks
 - Post-exploitation activity
-

Why Running as SYSTEM Matters

Many Windows services execute with:

NT AUTHORITY\SYSTEM

privileges.

SYSTEM is one of the highest privilege levels on Windows systems and provides attackers with:

- Full system access
- Access to sensitive files
- Ability to disable security controls
- Credential access opportunities
- Greater persistence capabilities

Malicious services running as SYSTEM significantly increase attack severity.

Event ID 7045 Investigation

Purpose

Windows Event ID 7045 records service installation activity.

This event helps analysts identify:

- New services
 - Persistence attempts
 - Suspicious service execution
 - Malware installation behavior
 - Unauthorized service creation
-

SPL Query Used

index=wineventlog EventCode=7045

Investigation Findings

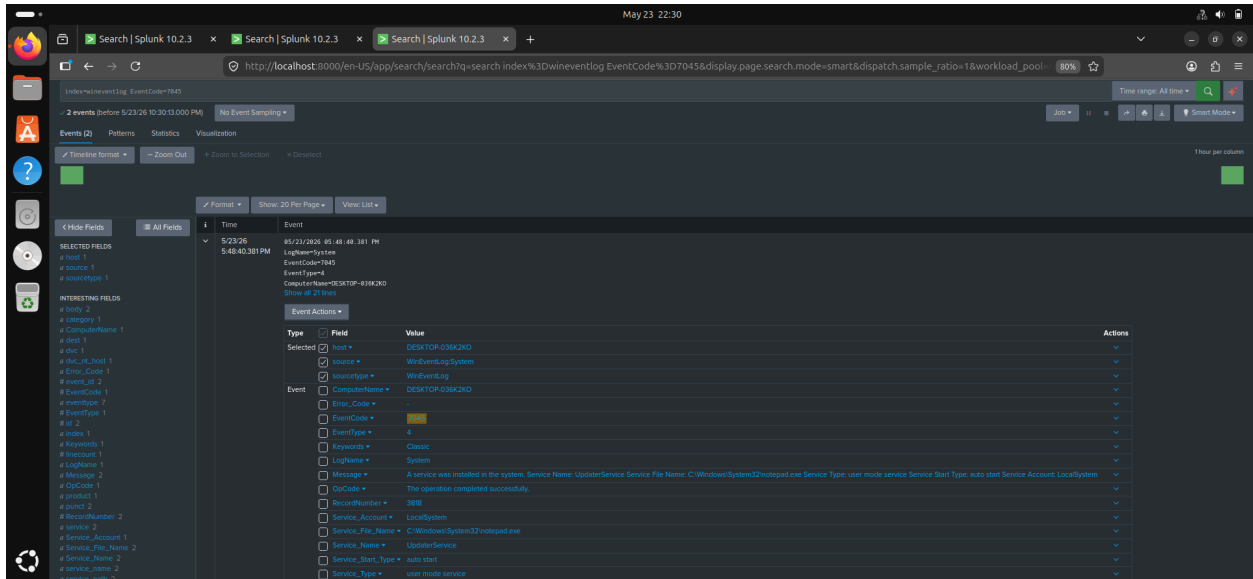
The investigation revealed:

- New service installation activity
- ServiceName visibility
- ImagePath visibility
- StartType configuration
- Service account information

Important fields analyzed included:

- ServiceName
 - ImagePath
 - ServiceType
 - StartType
 - AccountName
-

Screenshot 2 — Event ID 7045 Detection in Splunk



Which Service Names Increase Suspicion

Several service naming patterns increase suspicion.

Examples include:

- Fake Microsoft-looking names
- Misspelled Windows services
- Generic names like:
 - Updater
 - WindowsUpdate
 - SecurityService
 - DefenderUpdate
- Randomized service names
- Services mimicking antivirus software

Attackers commonly use deceptive names to blend malicious services into legitimate system activity.

Which ImagePaths Worry Defenders

Defenders become highly suspicious when services execute from locations such as:

C:\Users\
C:\Temp\
AppData\
Downloads\

Other suspicious indicators include:

- PowerShell configured as a service
- LOLBins used as service executables
- Encoded PowerShell commands
- Unsigned executables
- Network-accessing payloads

Examples of concerning executables include:

powershell.exe
cmd.exe
rundll32.exe
mshta.exe
regsvr32.exe

These are commonly abused in real-world attacks.

Legitimate vs Malicious Service Creation

Legitimate services are commonly created by:

- Operating system components
- Enterprise software
- Security tools
- Backup solutions
- Monitoring platforms
- System administrators

Legitimate services typically:

- Use trusted file paths
- Are digitally signed
- Have recognizable names
- Match expected software behavior

Malicious services often:

- Use suspicious execution paths
 - Execute scripting engines
 - Use deceptive service names
 - Attempt outbound network communication
 - Operate from user-controlled directories
-

Why Event ID 7045 Alone Is Insufficient

Event ID 7045 alone is often insufficient because service creation by itself does not always indicate malicious behavior.

Many legitimate applications install services regularly.

SOC analysts must therefore correlate 7045 with additional telemetry such as:

- Process creation events (4688 or Sysmon Event ID 1)
- PowerShell activity
- Network connections
- File creation events
- Authentication logs
- Endpoint detection alerts

Correlation helps determine whether service creation is part of a broader attack chain.

Correlating Service Creation with Process Execution

To improve investigation accuracy, I correlated:

- Event ID 7045 (service creation)
- Event ID 4688 (process creation)

This helps reconstruct attacker behavior and determine whether the created service actually executed.

Correlation SPL Query

index=wineventlog (EventCode=7045 OR EventCode=4688)

Correlation Findings

The investigation revealed:

sc.exe
└─ service installation
└─ notepad.exe execution

This demonstrated how service persistence can lead directly to process execution activity.

Screenshot 3 — Full Event Correlation Investigation

The screenshot displays a Splunk search interface with the following details:

- Search Bar:** `index=wineventlog (EventCode=7045 OR EventCode=4688) ("*UpdaterService*" OR "*notepad.exe*" "%0A%0A" evi`
- Results:** 34 events (before 5/23/26 8:19:57:000 PM). The table shows a sequence of events where the `sc.exe` service is installed and then `notepad.exe` is executed.
- Table Columns:** Time, Computer, User, Process_Name, Command_Line, LogonId.
- Table Data:**

Time	Computer	User	Process_Name	Command_Line	LogonId
2020-05-21 17:54:32.871	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-21 17:54:32.886	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-21 18:31:25.413	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-21 18:31:25.419	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-21 18:47:37.654	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-21 18:47:37.658	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\sc.exe	sc.exe /create /no minute /no 5 /in Updater /r notepad.exe	6x13bc
2020-05-22 05:12:00.988	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 05:12:00.992	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 11:41:22.137	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 11:41:22.219	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 12:11:06.331	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 12:11:06.336	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 13:06:45.961	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 13:06:45.963	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 16:02:44.843	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-22 16:02:44.848	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-23 09:27:06.114	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-23 09:27:06.119	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-23 17:56:48.318	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc
2020-05-23 17:56:48.319	DESKTOP-8J4K260	DESKTOP-8J4K260\prc...	C:\Windows\System32\notepad.exe	"notepad.exe"	6x13bc

Screenshot 4 — EventCode 1 Service Execution Correlation

The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index=winlog (EventCode=1 OR EventCode=4088) ("notepad.exe" OR "notepad.exe") | eval Process_Name=coalesce(Image, NewProcessName) | search Process_Name="C:\\Windows\\System32\\cmd.exe" | eval Command_Line=coalesce(CommandLine, ProcessCommandLine)`. The search results show a single event for EventCode 1, dated 5/23/26 5:48:40.339 PM. The event details are displayed in a table format, showing fields like host, source, sourcetype, channel, command_line, and parent processes. The event actions table is also visible, showing fields like host, source, sourcetype, channel, command_line, and parent processes.

Type	Field	Value	Actions
Selected	host	DESKTOP-030K2XO	
	source	XmWinEventLog Microsoft-Windows-Symon/Operational	
	sourcetype	XmWinEventLog	
Event	Channel	Microsoft-Windows-Symon/Operational	
	Command_Line	sc create UpdateService binPath="C:\\Windows\\System32\\notepad.exe" start= auto	
	Command_Line	sc create UpdateService binPath="C:\\Windows\\System32\\notepad.exe" start= auto	
	Company	Microsoft Corporation	
	Computer	DESKTOP-030K2XO	
	CurrentDirectory	C:\\Windows\\System32\\	
	Description	Service Control Manager Configuration Tool	

How Defenders Reduce Noise

Because service creation occurs frequently in enterprise environments, defenders reduce false positives by analyzing:

- Known-good service baselines
- Trusted software publishers
- Standard installation behavior
- Service naming conventions
- Execution paths
- Digital signatures
- Parent processes
- Historical service activity

Detection tuning is critical to avoid excessive alert fatigue.

Skills and Knowledge Gained

Through this project, I gained practical experience in:

- Windows service persistence detection
 - Event ID 7045 analysis
 - Service execution monitoring
 - Splunk SPL searches
 - Threat hunting concepts
 - Detection correlation
 - Persistence investigation
 - SOC investigation workflows
-

Detection Relevance

Service monitoring is highly important because services are one of the most abused persistence and execution mechanisms in Windows environments.

Monitoring service creation helps defenders identify:

- Malware persistence
- Unauthorized execution
- Privileged attacker activity
- LOLBin abuse
- Long-term compromise attempts

Service persistence detection is a critical SOC analyst skill.

Challenges Encountered

Some challenges encountered during the lab included:

- Understanding service telemetry
- Interpreting Event ID 7045 fields
- Distinguishing legitimate vs malicious services

- Understanding privileged execution
- Correlating service creation with process execution

Resolving these challenges improved my understanding of Windows persistence mechanisms and SOC investigation workflows.

Future Improvements

Future improvements planned for this lab include:

- Detecting malicious PowerShell services
 - Monitoring suspicious ImagePaths
 - Correlating services with network activity
 - Investigating lateral movement behavior
 - Simulating ransomware execution chains
 - Building service persistence dashboards
-

Conclusion

This lab improved my understanding of how attackers abuse Windows services for persistence and privileged execution and how SOC analysts investigate this activity using Splunk.

By analyzing Event ID 7045 and correlating it with process execution telemetry, I gained practical experience in:

- Persistence detection
- Service monitoring
- Threat hunting
- Event correlation
- Detection tuning
- SOC investigation workflows

This project strengthened my understanding of how Windows services are abused in real-world attacks and how defenders detect suspicious service activity in enterprise environments.

Author

Precious Anyanwu

Cybersecurity Learner | SOC Analyst Path | Splunk SIEM Practice